

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

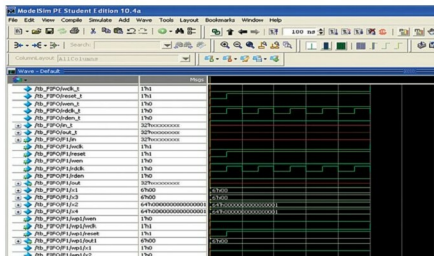


Fig. 13: Simulation window

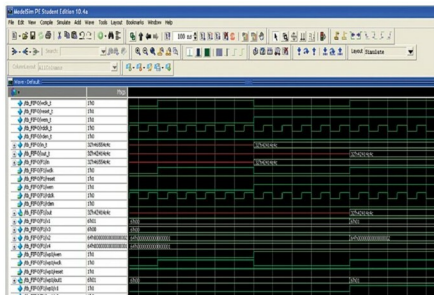


Fig. 14: Wave window

The final workspace window is shown in Fig. 7.

Compiling/debugging project files

1. Select Compile→Compile All options.
2. The compilation result is shown on the main window. A green tick is shown against each file name, which means there are no errors in the project (Fig. 8).

Simulating FIFO design

1. Click on Library menu from the main window and then click on

plus (+) sign next to the work library. You should see the name FIFO that we have just compiled (Fig. 9).

2. Now in work library, select tb_FIFO and click on OK (Fig. 10). This will open sim-Default window as shown in Fig. 11.
3. Go to Add→To Wave→All items in design options.
4. Select the signals that you want to monitor for simulation purposes. Select these as shown in Fig. 12.
5. Selected signals are the test bench to verify FIFO functional

behavior. We are now ready to simulate our design by clicking Run from Simulate menu bar as shown in Fig. 13.

Now, click on Zoom Full from Wave window. Your simulation output waveform will be as shown in Fig. 14. Note that, we have already provided 32-bit input values to all 64 address locations. You can check the first data as 32'h42_41_4C_4C in the first location on this screenshot. By scrolling the cursor towards right, you can see the data in the last location as 32'h41_52_54_49.

Thus, the result of FIFO design is verified from this output waveform.

Future applications

FIFO logic described here can be further refined to make more advanced-level projects. For example, in system on chip designs, there are components that often run on different clocks. So to pass data from one component to another, we need an asynchronous FIFO.

Sometimes, even if the source and request sides are controlled by the same clock signal, FIFO is needed. This is to match the throughputs of the source to the request. For example, the source may be supplying data at the rate at which the request cannot handle or, in other case, may be placing requests for data at a rate at which source cannot supply. Here, a synchronous FIFO is used that acts as an elastic buffer. **EFY**



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